Physics Basics: Introduction to Motion

Chapter 1: What is Motion?

Motion is the change of position of an object over time.
Speed is the rate of change of distance.
Velocity is the rate of change of displacement (includes direction).

Example: A car traveling at 60 km/h north is moving with:

- Speed: 60 km/h
- Velocity: 60 km/h north

Key Concepts:

1. Distance: Total path traveled (scalar)
2. Displacement: Change in position (vector)
3. Average Speed: Total distance / Total time
4. Average Velocity: Total displacement / Total time

Chapter 2: Understanding Acceleration

Acceleration is the rate of change of velocity over time.

Formula: $a = (v - v_0) / t$

Where:

- a = acceleration
- v = final velocity
- v_0 = initial velocity
- t = time interval

Examples of Acceleration:

1. A car speeding up from 0 to 60 km/h
2. A car slowing down (deceleration)
3. A car turning at constant speed (direction change)

Free Fall: Objects falling under gravity with $a = 9.8\ m/s^2$

Physics Basics: Newton's Laws

Chapter 3: Newton's Three Laws of Motion

First Law: An object at rest stays at rest, and an object in motion stays in motion unless acted upon by an external force.

Second Law: Force equals mass times acceleration ($F = ma$)

Third Law: For every action, there is an equal and opposite reaction.

Applications:

- Seat belts: Keep you moving forward (first law)
- Rockets: Push down to go up (third law)
- Car engines: Generate force to accelerate (second law)

These laws form the foundation of classical mechanics and are used to analyze motion in everyday situations.